



Value Add of Addax Composite drive shafts

Cooling Tower drive shaft details: DBSE: 4600mm Overall Length : 4700mm Motor Rating: 137 kW	Steel Shaft: Diameter: 10 inch Weight: 150 kg approx. Rexnord Addax - Composite Carbon Fiber shaft: OD: 8 inch Weight: 27 kg	
Benefits to End User / Owner		
Advantages	Benefits	Explanation
No corrosion	No Recurring Capital Costs	Addax drive shaft is made of corrosion free composite center member and flexible elements (non metallic) and SS316 (corrosion resistant) motor and Gear box side Hubs. In fact it is ideal for Coastal and Salt water applications. These shafts require almost no maintenance. It is deigned to last the lifetime of the Cooling tower and hence there are NO RECURRING Capital costs involved with replacement of entire shaft Steel shafts have a life of 2 years (Coastal) to 10 years (No Coastal) depending on environmental conditions and maintenance frequency. This will require replacement of shaft 2-4 times during lifetime of Cooling Tower.
Part replacement possible. Entire assembly need not be replaced.	Reduced recurring capital costs and inventory cost.	A steel shaft is assembly balanced. If a hub (coupling) is degraded due to corrosion or damage, the entire shaft assembly has to be replaced as inserting a new part will cause imbalance in the system. In an Addax shaft due to light weight nature , precision manufacturing of the shaft and tight tolerances of the hubs, only the center member is balanced. There is no requirement of assembly balance. Hence if a hub or flex element has to be replaced for any reason, only that part can be replaced without causing any imbalance in the system and hence not requiring the entire assembly to be replaced.
Reduced INVENTORY required due to interchangeability of parts	Cost savings	In an Addax assembly, a hub or flex element of a particular part no. can replace any hub or flex element in any shaft sharing the same part no. This results in reduced INVENTORY required. In case of steel, an entire drive shaft assembly has to be retained as inventory as parts cannot be used interchangeably

Reduced DOWN TIME during maintenance	3-4 hrs / shaft	During maintenance of Gear Box, drive shaft will have to be uninstalled and reinstalled. Ease of handling of light weight Addax composite shaft results in saving of time. Drop Down feature ensures no realignment is required. Only 3 bolts per end have to be removed to unistall shaft. The Motor hub and Gear Box side Hub remain in place and the shaft can very easily be removed and put back without disturbing the system.
Higher End equipment life	Less Down Time Low cost of ownership	Light weight, high misalihnment capacity Addax composite shaft results in longer life of end bearings. Zero vibration ensures longer life of Gear Box and motor. Weight of steel shaft and vibrations transmitted may cause catastrophic failure of bearings, Gear Box and motor resulting in Down time and capital cost.
No vibrations		Vibrations are caused by the following: i. Misalignment in the system ii. Corrosion cauing mass to be reduced unevenly iii. Improper installation Intensity of Vibration is directly proportional to Unbalanced mass. Addax Composite drive shaft is a low mass , corrosion resistant, precision manufactured, factory balanced drive shaft, that retains its straightness and dimensional stability. It offers a misalignment (absorbing) Capicity of 1 degree per end, which is 3 times the industry standard. In a steel drive shaft, the pin and bush arrangement tends to degrade rapidly resulting poor capacity to absorb misalignment in the system. Metal parts corrode faster resuting in uneven mass distribution. Self sagging due to weight, loss of straightness due to ductile nature, uneven wall thickness due to use of non- structural grade steel lead to unbalanced mass Higher the material density, greater is the possibility of unbalanced mass. Slight error during re-installtion during maintenance will cause vibrations. Therefore steel shaft results in greater vibration than light weight Addax Composite shafts.
No frequent changing of parts	Reduced down time	CTI confirms that the Rubber bushes in the Pin and Bush type flexible arrangement tend to get degraded quickly in a wet, humid and warm environment present in a cooling tower. In addition, chlorine used in CT water tends to harden the rubber causing it to loose its misalignment tolerating capacity. This requires that frequent changing of rubber bushings Addax composite flex elements have a min life of 5 years . Therefore system can function untouched and maintenance free for years
No multiple parts to be changed	Reduced down time	Addax Flex element is SINGLE part that can be easily replaced even without removing shaft from position. In a Pin and Bush arrangement there are MULTIPLE PARTS (at least 8 bushes on each side) that need to be routinely be replaced.

Benefits to Contractor		
Advantages	Benefits	Explanation
Installation time saved	2-3 hrs/ shaft	Ease of handling a light weight Addax composite drive shaft ensures that in comparison to a heavy steel shaft the time is saved in the following operations: 1. Lifting from ground to top of Cooling tower 2. Installation between motor and gear box 3. Alignment.
	Min:75-100 hrs saved/ 48 shafts	If there is a single team doing installation then ADDITIONAL DAYS required : a) 8 hr shift : 9 to 12 days b) 12 hrs shift: 6 to 8 days c) 24 hrs shift: 3 to 4 days
	0.5% of project cost	7 day (1 week) delay attracts a 0.5 % of project cost as LD. 1 in 3 projects come into the critical phase where there may
Reduces accidents	Safety maintained. No stoppage in work	Handling a light weigh 4.7 m shaft reduces the possibility of accidents, damage to shaft and structure.
Onsite balancing not required	3-10 hrs / shaft saved	Addax is precision manufactured and factor balanced . The high specific stiffness and strength of carbon fiber ensures shaft retains straightness and is dimensionally stable. Hence, no onsite balancing is required.
	Down-time & Cost of balancing saved	Onsite balancing will take between 3 to 10 hrs based on severity of imbalance, for 2 trained technicians with specialized machinery / equipment. In house cost would be Rs 5000 - 7000. If Imbalance is severe it will require machine balancing. This will involve transportation to and from the project site to the nearest balancing facility which may take between 4 to 20 days based on remoteness of site.

No Vibrations	No performance related complaints	Vibration due to material property: Vibrations are greater in steel shaft as compared to Addax composite shaft.
	Alignment error during installation can be accommodated without causing vibration.	Vibration due to improper installation: Vibration readings due to 0.02mm error in alignment: Steel shaft : 7-8 mm/s Addax Composite shaft: 3-4 mm/s
Critical Speed issues for long shaft are solved	No catastrophic failure due to resonance & vibration.	Critical speed formula, $N_{cr} = C \cdot (E \cdot I \cdot g / (w \cdot L^4)) \cdot .5$. Critical speed is inversely proportional to mass & length. For 4.7m steel shaft, with mass & length increasing critical speed drops closer to operating speed resulting in resonance & failure. For 4.7m Addax shaft, because of lower mass, critical speed is safely above operating speed.
Guaranteed trouble free operation	No onsite maintenance visits required. 10% Performance Guarantee not held up	Using Addax Composite drive shafts ensures trouble free operations for many years (lifetime of Cooling Tower) Any Problems due to Steel shafts, eg misalignment, vibration may lead to repeated maintenance visits to site & in some severe cases may lead to performance guarantee amounting to 10% of project cost being held up.
Satisfaction of using cutting edge technology from a world class company	Recognition in Industry for introducing Advanced products	Distinct advantage over contractors using steel as they will be considered regressive. Industry will appreciate your efforts for choosing to use ONLY WORLD CLASS products by opting to use Rexnord Addax composite drive shafts . Rexnord is the world's largest coupling company and a respected name in Power transmission for the past 130 years. Rexnord Addax pioneered Composite CT drive shafts & is the market leader globally. It has over 80% market share in the US and 60% globally. It is recognised by NTPC, EIL, TCE, DCPL, Bechtel and others and is installed in projects at NTPC, Adani, Reliance, Essar, Tata, GMR. Rexnord has an install base of over 50,000 drive shafts globally & over 500 in India, the largest among any composite manufacturer. With composite drive shafts being accepted as preferred MOC, contractors would be expected to offer the lighter, corrosion carbon fiber drive shafts.